

## Genome-Wide Association Study of T2D: Comparative Analysis in East Asian Populations

### Introduction

Type 2 Diabetes (T2D) is a chronic metabolic disease characterized by hyperglycemia and insulin resistance, affecting millions globally<sup>1</sup>. Despite the highly polygenic nature of type 2 diabetes, recent studies have indicated the effects of individual loci differs significantly across ancestral populations. Many Genome-Wide Association Studies (GWAS) have been conducted on European cohorts, potentially overlooking high-impact variants that are more prevalent in non-European populations. Among East Asian populations, type 2 diabetes is more frequently associated with defects in pancreatic beta-cell insulin secretion, rather than the insulin resistance commonly seen in Western groups<sup>2</sup>. This physiological difference highlights the importance of identifying population-specific genetic drivers. Among these, the *KCNQ1* locus on Chromosome 11 has been identified as a key genomic marker<sup>3</sup>.

The *KCNQ1* gene encodes a voltage-gated potassium channel that is essential for maintaining electrical excitability in multiple tissues, including the heart and pancreas<sup>4</sup>. Genetic variation in this gene has been linked to reduced insulin release. While previous multi-ancestry meta-analyses, such as Zhao et al. (2017), identified the *KCNQ1* locus as a significant susceptibility signal, complete understanding of its impact within East Asian populations required higher-resolution, ancestry-specific data. Thus, this study aims to clarify the broader significance of the *KCNQ1* gene using summary statistics from the Spracklen et al. (2020) meta-analysis, the largest East Asian T2D study to date. The analysis demonstrates that *KCNQ1* is a dominant genetic contributor to T2D in East Asian populations, providing insight into population-specific disease mechanisms.

### Methods

Data for this study was obtained from the GWAS Catalog, primarily focusing on the Spracklen et al. (2020) meta-analysis of Type 2 Diabetes (GCST010118). This dataset represents a comprehensive study of 77,418 cases and 356,122 controls of East Asian ancestry. Initial exploratory analysis was also conducted using summary statistics from Zhao et al. (2017) to confirm the presence of established susceptibility signals and to provide a comparative baseline for the subsequent analysis.

The analyses were conducted in R. The datasets were filtered to include only autosomal variants (Chromosomes 1-22). Missing values for genomic coordinates or p-values were removed to ensure statistical integrity. Whole-genome Manhattan plots were generated using the qqman package. P-values were transformed to a  $-\log_{10}$  scale to visualize the distribution of association signals across the genome. The lead variant at the *KCNQ1* locus was identified by filtering for the highest significance within the Chromosome 11 regional coordinates. Effect sizes were analyzed using odds ratios (OR) to determine the magnitude of risk associated with the effect allele. A significance threshold of  $P < 5 \times 10^{-8}$  was applied to identify genome-wide significant loci. Additionally, Quantile-Quantile (Q-Q) plots were constructed to compare the distribution of observed p-values against the expected null distribution.

Results

The genome-wide analysis of the Spracklen and Zhao et al. datasets revealed highly significant genetic associations for T2D in East Asian populations. Initial exploratory results from the Zhao et al. cohort identified a strong susceptibility signal at the KCNQ1 locus on Chromosome 11 (Fig. 1b), with variant rs2237892 as the lead variant ( $P = 2.0 \times 10^{-63}$ ). However, the primary analysis using the Spracklen et al. East Asian meta-analysis ( $N = 433,540$ ) provided more reliable association signals. As shown in the Manhattan plot (Fig. 1a), multiple loci surpassed the genome-wide significance threshold ( $P < 5 \times 10^{-8}$ ). The most prominent association is located on chromosome 11 at the KCNQ1 locus, with the lead variant rs2237897 reaching a significant p-value of  $2.0 \times 10^{-245}$ . Demonstrated in Table 1, this variant is associated with an odds ratio of 1.28, indicating that each additional copy of the risk allele is associated with a 28% increase in the odds of developing Type 2 Diabetes.

Table 1: Top Loci for Type 2 Diabetes (Spracklen et al., 2020)

| SNP (rsID) | Chr | Locus/Gene      | P-value  | Odds Ratio (OR) |
|------------|-----|-----------------|----------|-----------------|
| rs2237897  | 11  | KCNQ1           | 2.0e-245 | 1.28            |
| rs9350271  | 6   | CDKAL1          | 5.0e-183 | 1.21            |
| rs10965248 | 9   | CDKN2A, B       | 4.0e-164 | 1.20            |
| rs2233580  | 7   | PAX4            | 3.0e-132 | 1.34            |
| rs61342118 | 7   | PAX4, SND1, LEP | 5.0e-105 | 1.34            |
| rs35906730 | 10  | HHEX, IDE       | 1.0e-71  | 1.15            |
| rs13266634 | 8   | SLC30A8         | 4.0e-67  | 1.12            |
| rs13092876 | 3   | IGF2BP2         | 2.0e-66  | 1.13            |
| rs7901695  | 10  | TCF7L2          | 8.0e-62  | 1.32            |
| rs11257657 | 10  | CDC123, CAMK1D  | 1.0e-61  | 1.12            |

Source: GWAS Catalog GCST010118

The Q-Q plot (Fig. 2) shows that the majority of observed p-values follow the expected null distribution, with a sharp upward curve at the tail. This pattern confirms that the observed associations at the KCNQ1 locus are true biological signals and that genomic inflation from population stratification was controlled for.

Discussion

The extreme statistical significance of the KCNQ1 locus ( $P = 2.0 \times 10^{-245}$ ) in this analysis provides a definitive genomic profile of Type 2 Diabetes risk within East Asian populations. The results are built on the foundational work of Zhao et al. (2017), which identified KCNQ1 as a lead susceptibility locus ( $P = 2.0 \times 10^{-63}$ ) in a smaller multi-ancestry cohort centered around variant rs2237892. Utilizing the Spracklen et al. (2020) dataset in a subsequent analysis, focus was exclusively placed on East Asian ancestry, while the sample size ( $N = 433,540$ ) tripled. From this, we observed an increase in power and a shift in the lead signal from rs2237892 to rs2237897. This shift exemplifies the concept of fine-mapping. As  $N$  increases, our ability to distinguish the true causal variant from neighbors in high linkage disequilibrium (LD) improves significantly. While the 2017 study was able to capture the gene’s signal, the 2020 data provide the power necessary to identify the primary genomic driver. These results suggest that we cannot rely on European-centric data to treat global populations; ancestry-specific research is necessary for building equitable and accurate precision medicine tools.

## References

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## Supplementary Materials

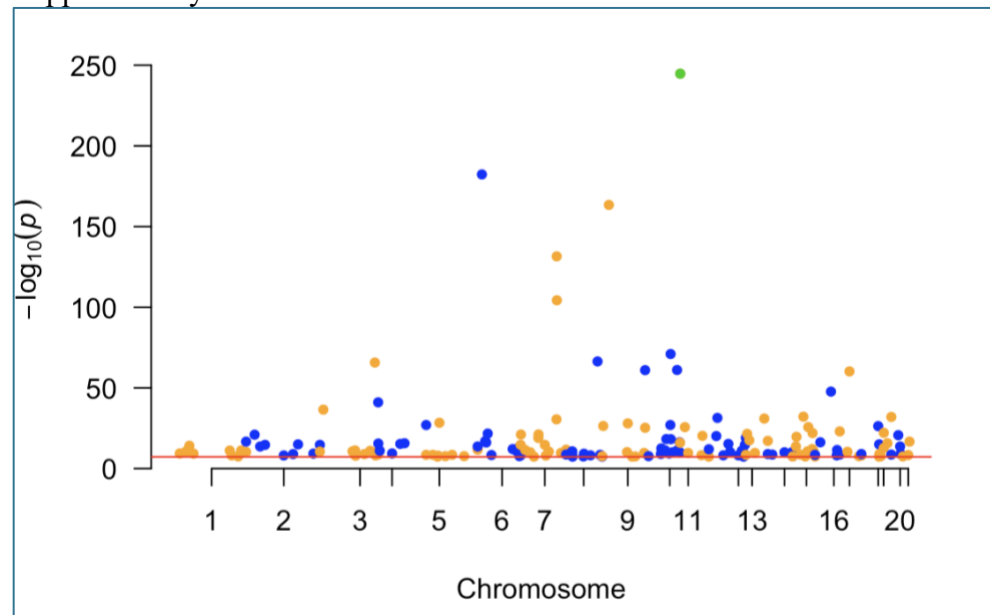


Figure 1a. Manhattan Plot of East Asian Type 2 Diabetes: Spracklen et al. (2020)

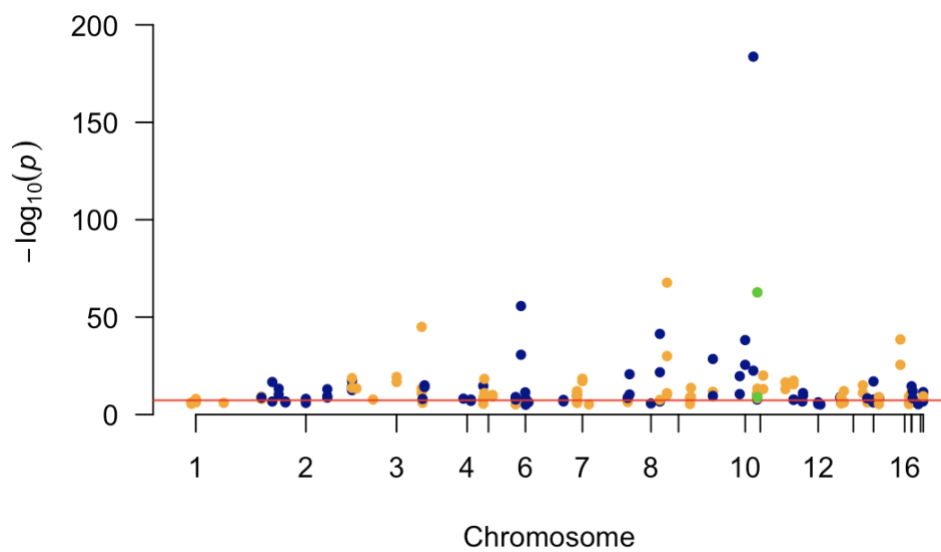


Figure 1b. Manhattan Plot of Type 2 Diabetes: Zhao et al. (2017)

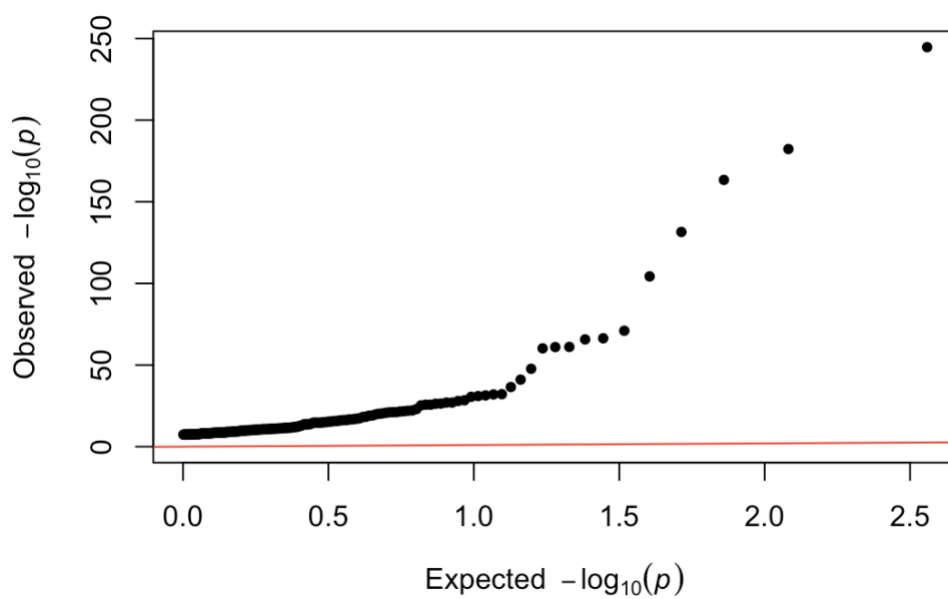


Figure 2. Q-Q plot for T2D on Spracklen et al. (2020)